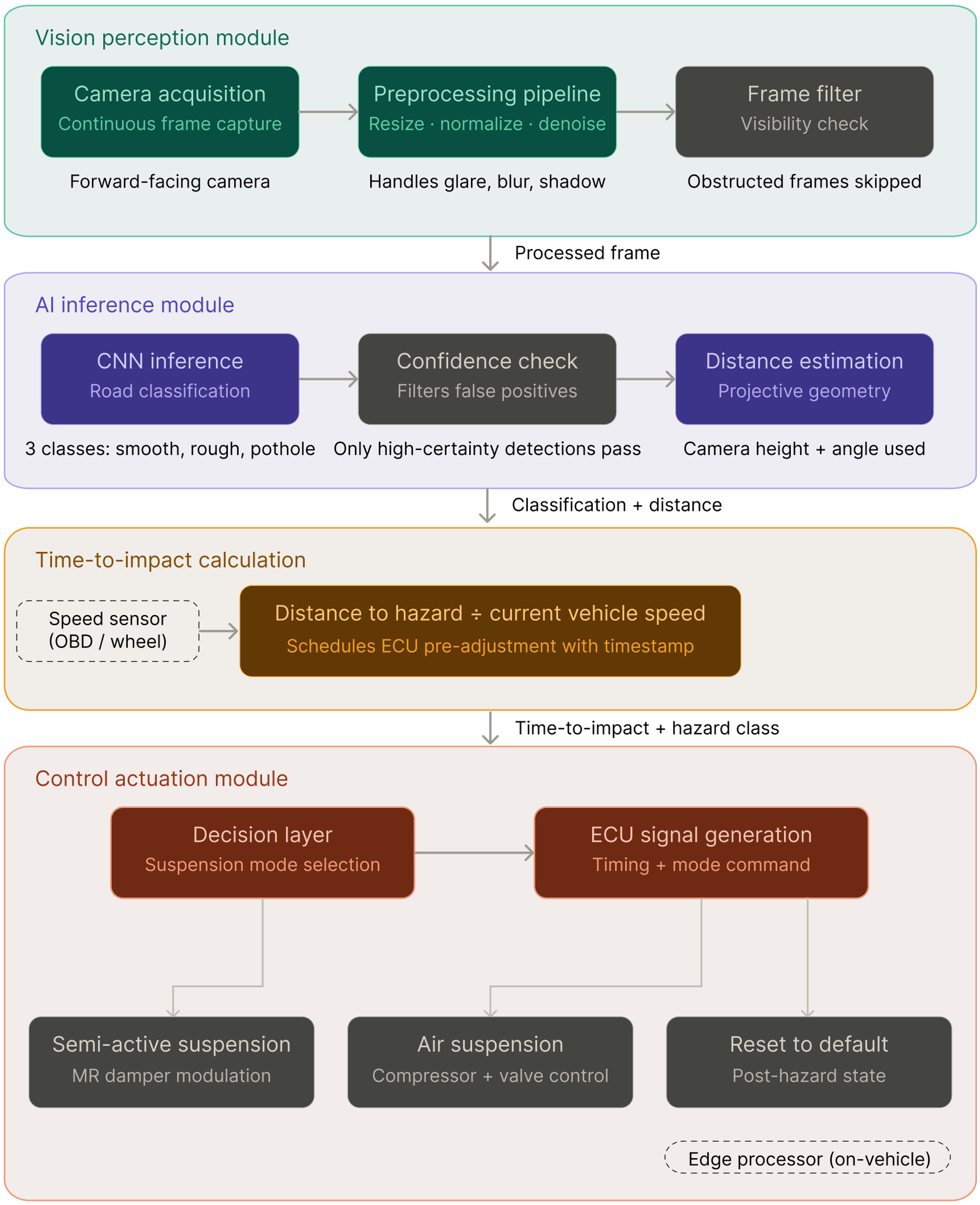

Vision perception module
Camera acquisition
Continuous frame capture
Forward-facing camera
Preprocessing pipeline
Resize · normalize · denoise
Handles glare, blur, shadow
Frame filter
Visibility check
Obstructed frames skipped
Processed frame
AI inference module
CNN inference
Road classification
3 classes: smooth, rough, pothole
Confidence check
Filters false positives
Only high-certainty detections pass
Distance estimation
Projective geometry
Camera height + angle used
Classification + distance
Time-to-impact calculation
Speed sensor (OBD / wheel)
Distance to hazard ÷ current vehicle speed
Schedules ECU pre-adjustment with timestamp
Time-to-impact + hazard class
Control actuation module
Decision layer
Suspension mode selection
ECU signal generation
Timing + mode command
Semi-active suspension
MR damper modulation
Air suspension
Compressor + valve control
Reset to default
Post-hazard state
Edge processor (on-vehicle)